

Five Specifications, One Horse Race: The Role of Credit versus Trade in Crisis Transmission

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I. Introduction

When financial crises strike, they are rarely contained within borders. A shock originating in one country can devastate economies thousands of miles away, even those with no meaningful direct exposure to the epicentre. The Global Financial Crisis provides a way to illustrate this. Following the collapse of the US subprime mortgage market in 2007, the crisis triggered a domino effect that culminated in a global banking panic with the failure of Lehman Brothers in 2008. This correction into the housing market in the United States escalated into the deepest global recession since the Great Depression.

Yet all countries did not suffer equally: some economies contracted sharply while others weathered the storm. Japan stands out as a case in point. Despite holding minimal direct exposure to subprime assets in the US, Japan was amongst those advanced economies hardest hit by US subprimes after 2008. The mechanism was not trade or toxic assets: it was financial linkages operating through an unexpected channel. Japanese banks held nearly \$190 billion in equities, and when foreign investors sold approximately one-third of those to meet their own liquidity needs elsewhere, the resulting equity collapse devastated bank balance sheets. This generated valuation losses several times greater than Japan's actual subprime losses (IMF, 2008).

What determines who gets hit hardest thus remains one of the central questions in international macroeconomics. So in late 2007, when the Global Financial Crisis occurred, it raised a fundamental question that remains incompletely answered: through which channels does a financial shock of this magnitude travel across countries? Two candidates dominate the debate:

credit - which operates through the depth of domestic credit markets and the vulnerability of balance sheet deteriorations - and trade - which operates through collapsing export demand and supply chain disruptions. Despite the existence of a rich literature on both, few studies put them in direct competition. Thereby, it becomes imperative to look at them side by side through a horse-race specification.

By tracing how pre-crisis financial and trade exposure shaped growth outcomes across countries during the Global Financial Crisis, this thesis examines which channel dominated transmission, leading to the research question:

Did pre-crisis credit exposure or trade openness better explain differential GDP growth outcomes across countries during the Global Financial Crisis?

The thesis' hypothesis follows from the financial accelerator literature: countries with deeper domestic credit markets entering the crisis should have systematically experienced larger contractions, while trade openness exhibits weaker effects.

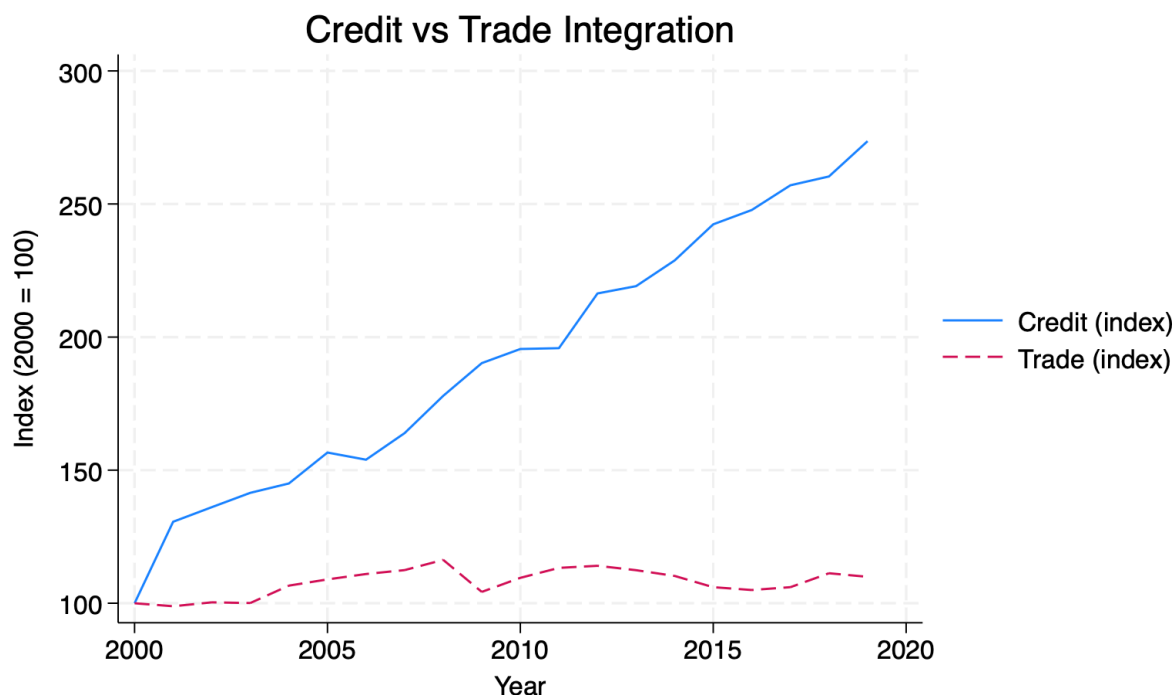
To address this, the paper uses panel data from Lei, Li, and Mei (2023), covering 130 economies over 2000–2019, employing a country fixed effects horse-race specification that allows both channels to compete simultaneously. The findings of this are validated through five distinct specifications to test whether one channel dominates the other, regardless of the methodology. For the baseline specification, a pooled OLS model is used. After that, a country fixed effects (FE) model is used, which allows the regression framework to control for time-invariant country

characteristics by comparing each country to itself over time. The framework is extended by using a two-way fixed effects (TWFE) model, which incorporates both country and time fixed effects to absorb global shocks common to all countries. Given that countries enter formal financial crises at different times, a staggered differences-in-differences (DiD) estimator is used. This is done to account for the fact that treatment - the onset of crisis - occurs in different years across countries. Finally, a probit model is employed to estimate the probability of the incidence of a crisis - to examine whether credit or trade exposure better explains which countries entered a formal crisis between 2007 and 2009.

Throughout the specifications, the credit channel consistently dominates. Trade openness is statistically insignificant across all specifications and appears protective in the staggered framework, which means that more open economies experience smaller contractions relatively. The probit results reinforce this pattern: credit exposure significantly raises the probability of entering a formal crisis; trade exposure does not.

This question is not merely academic. It remains imperative as identifying the dominant transmission channel has direct implications for the way in which countries design macroprudential policy to safeguard against vulnerabilities. In an increasingly interconnected global economy - where financial integration has accelerated since 2008 even as though trade linkages have plateaued, as shown in figure 1 - resolving this question becomes even more important to prepare for when the next shock arrives.

Figure 1. Evolution of Global Credit and Trade (2000-2019)



Notes: Indices set to 100 in 2000. Credit index represents the mean domestic credit-to-GDP ratio; trade index represents the mean trade share of GDP. Credit deepened over 2000–2019; whereas, trade peaked in 2008 and stagnated thereafter.

This paper contributes to the literature by shifting the lens from single-channel narratives to a direct competition between credit and trade. It tests both simultaneously across three methodologically distinct specifications - each carrying different identifying assumptions - to examine whether the dominance of credit over trade holds regardless of how transmission is measured. In doing so, it moves beyond the descriptive to offer systematic evidence on which vulnerabilities policymakers should prioritise when building resilience against global financial shocks.

II. Literature Review

What is a financial crisis? At its core, a financial crisis refers to a severe disruption in the functioning of financial markets, which can result in declines in asset prices, widespread insolvency amongst financial institutions, and a significant contraction of economic activity (Reinhart and Rogoff, 2009). These crises can trigger failures that amplify initial shocks and lead to severe economic consequences.

There exists a wide array of literature covering not just the causes and the consequences, but also the historical elements of financial crises. One of these is *This Time is Different* by Reinhart and Rogoff, which looks at these events across different countries over eight centuries through a historical and quantitatively analytical look. The one common predecessor of these crises, according to the authors, remains excessive debt accumulation. Through systematic documentation of 66 countries over multiple centuries, they establish that financial crises follow consistent patterns, despite each generation's belief that "this time is different."

According to the authors, different crises can be classified into distinct types: banking crises occur when bank runs, failures, or large-scale government interventions threaten the stability of the financial system; sovereign debt crises involve a government defaulting on either external or domestic debt; currency crises are characterized by sharp exchange rate depreciations; and inflation crises occur when annual inflation exceeds 20 percent. It is important to note, however, that in our interconnected macro world, these crisis types rarely occur in isolation - banking crises precede sovereign debt crises, and both can coincide with currency crashes.

Reinhard and Rogoff also discuss the transmission mechanisms for these crises, particularly those linked to debt and leverage. They argue that large-scale debt accumulation by economic agents such as governments, corporations, or households can create fragility within the economic system. This makes economies vulnerable to sudden shifts in investor confidence, which is important especially when debt is short-term or externally financed, requiring continuous rollover in capital markets. Due to shifts in sentiments, a sudden stop in capital flows can trigger self-fulfilling panics as lenders withdraw funding.

The authors highlight the way in which asset price booms can amplify crisis dynamics, where debt fueled expansions can inflate housing and equity markets beyond sustainable levels. And when the bubbles burst, the consequences are felt across institutions and countries - transmitting distress through financial linkages rather than direct trade flows. This analogy brings us to the Global Financial Crisis. Reinhart and Rogoff termed it as the 'Second Great Contraction', which draws a parallel to the Great Depression as the first major global contraction. The Global Financial Crisis originated within the subprime mortgage market in the United States. While direct exposure to these mortgage assets was limited in many countries, it did not insulate them from the downturn. In this scenario, transmission occurred through securitized debt instruments that distributed subprime mortgage risk globally, through international banking linkages as institutions reassessed counterparty risk, and through sudden reversals in cross-border capital flows. In addition, as global economic activity contracted, trade transmission also operated through a decline in aggregate demand, as falling output led to reduced imports across economies. Several pre-crisis indicators that were identified by the authors were fundamentally financial rather than trade-based. The severity and speed of transmission through financial

channels, compared to the more gradual contraction in trade flows, suggest that credit linkages may have been the dominant transmission mechanism.

While Reinhart and Rogoff's framework establishes that credit channels should dominate trade channels in transmitting crises, their framework does not formalize the mechanisms by which these linkages amplify shocks. Bernanke and Gertler (1995) provide a formal theoretical framework that explains why and how financial shocks propagate through credit markets. By modeling balance sheets and bank lending channels, they clarify the mechanisms that can amplify shocks in highly leveraged economies, offering a microeconomic foundation for the patterns observed in historical crises. This framework helps bridge the gap between descriptive historical evidence and formal modeling of crisis transmission, informing subsequent empirical and theoretical work on the relative roles of trade and credit channels. According to them, credit markets are inherently imperfect due to information asymmetries, agency problems, and moral hazard. They argue that traditional views, which focus solely on interest rates and aggregate demand, are incomplete; thus, it is important to look at frictions within credit markets that create a powerful amplification mechanism, called 'the financial accelerator'.

Bernanke and Gertler (1995) show that credit market imperfections can amplify shocks through two channels: the balance sheet channel, where declines in borrowers' net worth reduce their spending and investment, and the bank lending channel, where impaired banks restrict credit even to otherwise sound borrowers. Both channels create self-reinforcing cycles that propagate financial distress across the economy. This framework is directly relevant to understanding crisis transmission: it explains how financial shocks can spread internationally through credit linkages,

independently of trade flows, highlighting the potential dominance of the credit channel over the trade channel in transmitting crises.

This framework helps explain why financial shocks have disproportionately large and persistent effects compared to other disturbances. The financial accelerator transforms a shock of given magnitude into a sequence of iteratively amplified contractions as credit constraints bind, loosen slightly during partial recovery, then bind again when new adverse information emerges. This mechanism suggests clear empirical predictions: economies with higher leverage - more debt relative to assets or income - should experience more severe contractions during financial crises as balance sheet deterioration triggers stronger accelerator effects.

The microeconomic foundations for how these credit market frictions transmit shocks across borders are formalised in Allen and Gale (2000). Their model shows that in normal times, banks hold deposits in institutions in other regions as an insurance arrangement against local liquidity shocks, allowing the interbank market to reallocate liquidity efficiently. When an unexpected aggregate shock occurs, however, these same linkages become the primary vector of contagion - to meet withdrawal demands, banks liquidate their claims on other institutions, spreading financial distress through a chain of forced liquidations. Crucially, Allen and Gale demonstrate that this contagion mechanism operates entirely through overlapping claims in the banking sector, even in the complete absence of trade or currency linkages. This makes their model the theoretical pure type for the credit channel hypothesis - financial crises can propagate solely through balance sheet interconnections without requiring any real economy transmission. The structure of the network matters critically: incomplete financial networks, where institutions are

connected to only a limited number of counterparties, are most fragile because a shock to one node hits its neighbours with full force rather than being distributed across the system. This insight applies directly to the GFC, in which specific bilateral exposures to US shadow banking products created concentrated transmission pathways that allowed the initial subprime shock to propagate rapidly to financial systems across the world.

Krugman (1999) extends the logic of the balance sheet channel to the international context, showing how credit disruptions can propagate across borders even when trade linkages are weak. In the aftermath of the Asian Financial Crisis, Krugman seeks to explain the "stunning" contagion between economies that lacked significant direct trade or financial connections with crisis epicenters. This analysis directly challenges the adequacy of trade-based explanations for the transmission of crises.

One of these explanations is the transfer problem - the sudden need for a crisis-hit country to shift its current account from deficit to surplus. Within traditional trade models, this adjustment occurs through the depreciation of a currency - to make exports cheaper and imports more expensive - to gradually rebalance the current account. However, Krugman shows that when countries carry substantial foreign-currency-denominated debt, credit and balance sheet effects dominate.

The mechanism operates through a "circular loop" between credit markets and the real economy. Capital outflows are triggered by a loss of creditor confidence, which forces the value of the currency to depreciate further. If countries possess large foreign-currency debt, this depreciation

mechanically increases the value of their domestic-currency debt obligations. As a result, this deteriorates the balance sheets of banks and firms which validates the creditors' initial pessimism, triggering further capital withdrawal and depreciation.

The mechanism forms a “circular loop” between credit markets and the real economy. Due to the fact that many countries are linked through cross-border lending and foreign-currency exposures, this process transmits financial shocks internationally, even in the absence of strong trade links. A growing body of empirical research examines these mechanisms in the context of the Global Financial Crisis. Two relevant contributions are made by Kalemli-Ozcan, Papaioannou, and Perri (2013) and Lane and Milesi-Ferretti (2011), which provide evidence that cross-border financial linkages, rather than trade exposures, served as the primary channels for transmitting and amplifying the crisis across countries.

Kalemli-Ozcan et al. (2013) analyze bilateral banking linkages using Bank for International Settlements (BIS) data on cross-border banking positions between country pairs. Their empirical framework finds the link between business cycle synchronization to the intensity of pre-crisis banking integration, while looking into the effect of financial integration to differ between crisis and non-crisis periods. They find that before 2007, greater banking integration was associated with more divergent business cycles, which is consistent with international risk-sharing theories. During the crisis, however, this relationship reversed sharply: financially integrated countries exhibited significantly higher output co-movement, indicating that banking linkages had become conduits for shock transmission. The results remain robust even after controlling for bilateral

trade flows, which suggests that the observed synchronization was not due to the movement of trade flows.

While Kalemli-Ozcan et al. examine bilateral banking flows among a relatively small set of advanced economies, Lane and Milesi-Ferretti (2011) adopt a broader cross-country approach to explain variation in crisis severity. They regress average GDP growth between 2008 and 2009 on pre-crisis country characteristics to find that variables related to credit remain the strongest predictors of decline in output. Trade openness and export composition, in contrast, explain only a modest portion of cross-country variation in growth outcomes. Although trade exposure contributed to the global downturn, financial vulnerabilities proved far more powerful in explaining the depth of individual country recessions. Their evidence points to sudden stops in capital flows and balance sheet deterioration as the primary mechanisms driving domestic demand collapses.

Extending the credit channel evidence to a longer historical horizon, Schularick, and Taylor (2013) document across 14 advanced economies from 1870 to 2008 that more credit-intensive expansions are systematically followed by deeper recessions and slower recoveries. Their central finding - that financial crisis recessions leave real GDP per capita approximately 5 percent below peak after five years, compared to a full recovery within two years in normal recessions - establishes that the credit channel mechanism is not specific to the GFC but reflects a robust historical regularity spanning multiple centuries and crisis episodes. Using local projection methods, they further show that the severity of this relationship is increasing in the degree of pre-crisis leverage, such that countries with larger credit booms for instance the United States,

United Kingdom, Spain, and Ireland experienced substantially more sluggish recoveries than those entering the crisis with smaller balance sheets, such as Germany and Switzerland. Notably, their results remain significant even when conditioning on trade-related variables including the current account to GDP ratio, providing further evidence that financial factors operate as an independent driver of crisis severity rather than as a proxy for trade vulnerabilities. This paper builds on their findings by testing the same credit channel mechanism within a broader cross-section of 130 advanced and developing economies for the specific GFC episode, using a panel identification strategy that allows direct comparison of the credit and trade channels within a unified empirical framework.

Miranda-Agrippino and Rey (2020) extend the credit channel argument to a systemic level, documenting the existence of a Global Financial Cycle: a single common factor that explains over 20 percent of the variance in international risky asset prices, driven primarily by US monetary policy. By turning to the Bayesian VAR framework, they show that US monetary policy shocks transmit across borders through financial conditions, specifically through the leverage of global financial intermediaries and international credit flows, independently of trade-related demand channels. Even after controlling for global real economic activity, credit and bank leverage contract significantly in response to US monetary tightening, which suggests that financial spillovers operate as an independent mechanism that cannot be reduced to trade linkages.

What makes this particularly relevant for the horse-race design of this paper is the authors' finding on exchange rate regimes. Countries with floating exchange rates, which under the Mundellian trilemma should be insulated from foreign shocks, experience financial spillovers

just as severely as those with fixed rates. This is important because if trade were the primary transmission channel, exchange rate adjustment would provide a natural buffer. The finding that it does not directly foreshadow the results of this paper: that trade openness had no systematic effect on GFC growth outcomes once credit exposure is accounted for.

Taken together, these studies provide complementary evidence that credit market linkages dominated crisis transmission. The contributions suggest that vulnerabilities embedded in the financial systems determined how the U.S. financial shock transmitted internationally.

This thesis builds on their findings in different ways. Unlike Kalemli-Ozcan et al., which focuses exclusively on bilateral banking flows among advanced economies, this paper employs aggregate credit measures across a broader sample of 130 developing and advanced economies. Moreover, it explicitly compares credit and trade channels in a unified horse-race specification to test which mechanism explains transmission better. Moreover, it employs a panel framework with country fixed effects to go beyond the 2008-2009 period.

Although credit channels appear to dominate crisis propagation, several studies highlight that trade linkages can also transmit shocks internationally. It is, thus, imperative to understand the relative importance of trade versus credit channels, particularly during events like the Global Financial crisis - which involved a collapse of both international financial markets and led to an unprecedented contraction in world trade.

Trade-based transmission operates through declining export demand, supply-chain disruptions, and terms-of-trade effects. According to theory, trade-based transmission is contingent on the interconnectedness of countries through common demand and supply shocks. Obstfeld and Rogoff (2001) provide a foundational framework to explain how trade integration shapes business cycle co-movement. Recession in a specific country reduces aggregate income, which leads to a fall in demand for imports, transmitting negative demand shocks to trading partners. In a similar manner, supply disruptions propagate through input and output linkages between trading partners. These trade frictions, according to the authors, form part of the several international macroeconomic puzzles attributed to incompleteness within financial markets. In their framework, trade openness fundamentally determines the extent of international risk-sharing and macroeconomic synchronization, and trade and credit channels interact rather than operate independently.

The quantitative dominance of credit over trade as a transmission mechanism is shown directly by Obstfeld, Shambaugh, and Taylor (2010), who compare the speed and scale of credit-driven and trade-driven reserve drains during financial crises. Their analysis shows that even a severe trade deficit represents a reserve drain of only approximately 0.1 percent of GDP per week, which is a slow leak that economies can absorb gradually through standard current account adjustment. By contrast, a domestic banking crisis can generate capital outflows of 5 to 10 percent of GDP per week as depositors convert bank liabilities into foreign currency - this is termed as a double drain by the authors. This order-of-magnitude difference in the speed and scale of the two channels provides an intuitive explanation for why pre-crisis credit exposure should dominate pre-crisis trade openness as a predictor of GFC growth outcomes. Trade

linkages transmit shocks gradually through the real economy, while credit linkages can propagate shocks almost instantaneously through financial markets.

The GFC led to an extraordinary contraction in global trade. Levchenko, Lewis, and Tesar (2010) document that world trade fell by roughly 12 percent in 2009, exceeding the decline in global GDP by six times. This “Great Trade Collapse” initially appeared to signal a powerful trade transmission channel. However, upon closer examination, it suggested that the magnitude of the decline reflected primarily a synchronized global demand shock, amplified by the structure of international trade. Since traded goods are highly cyclical, declines in consumption and investment reduced trade flows disproportionately. When trying to test whether this was driven by a breakdown in trade finance, Levchenko et al. found little evidence that sectors more reliant on trade credit experienced larger contractions. Trade credit also witnessed a modest reduction, which corroborated their suggestion that financial distress affected trade mainly indirectly through reduced aggregate demand rather than through a direct finance–trade disruption. They also emphasized how structures of global production varied - with sectors producing input goods facing sharp declines, amplifying shocks along global supply chains. This resulted in a ‘trade wedge’, where reductions in final expenditure triggered multiple cross-border contractions in intermediate goods trade - and making trade fall substantially more than standard models predict.

According to Chor and Manova (2012), credit-constrained exporters reduced exports more sharply during the crisis, which shows the link between financial and trade channels.

To look at the distinction between trade linkages and credit growth, Lane and Milesi-Ferretti (2011) show that pre-crisis credit growth, leverage, and current account deficits are substantially

stronger predictors of 2008–2009 growth outcomes than trade openness through cross-country evidence. The distinction suggests that financial vulnerabilities within credit markets better explain differences in severity, while trade linkages account for synchronization of downturns across countries.

The literature establishes that the evidence does not conclusively demonstrate that trade constituted the primary transmission mechanism once financial fragilities are accounted for. Most studies examine trade and credit channels separately rather than directly, so the question of which channel dominated during the GFC remains open. Thereby, it is useful to turn to earlier empirical work that directly framed contagion as a finance-versus-trade question. A foundational contribution to this debate is the IMF Working Paper “Sources of Contagion: Finance or Trade?” by Van Rijckeghem and Weder (1999), which explicitly tests whether crises spread through trade competition or through shared financial exposures.

The paper analyzes the Mexican (1994), Thai (1997), and Russian (1998) crises, and the authors develop and empirically test the “common bank lender” hypothesis, which posits that crises propagate internationally when banks exposed to a ground-zero country curtail lending elsewhere to restore capital-adequacy ratios, satisfy margin calls, or reduce value-at-risk exposure. The paper uses probit regressions in which the dependent variable captures crisis incidence. It also constructs a measure of “competition for funds” that captures the extent to which countries share exposure to the same international banks. This variable serves as a proxy for financial interconnectedness and is directly contrasted with measures of bilateral trade linkages and third-market export competition, alongside standard macroeconomic controls. The

results show that exposure to a common lender is statistically significant with a higher probability of contagion, whereas trade linkages lose statistical significance once financial competition is included. For the episodes in Mexico and Russia, the findings suggest that financial retrenchment appears to have transmitted shocks even in the absence of substantial direct trade ties. The authors also acknowledge the high correlation between trade and financial linkages, noting that bank lending historically followed trading routes through trade finance, which complicates identification and may bias studies toward overstating trade effects.

Three critical gaps emerge from this literature. First, the horse-race question remains inadequately tested for the GFC using panel methods that can control for time-invariant country heterogeneity. While many studies examine credit or trade linkages separately, few directly compare them within a unified empirical framework that allows both channels to compete as explanations for crisis transmission. The studies that do compare channels rely primarily on cross-sectional variation, which cannot separate channel effects from confounding country characteristics.

Second, identification challenges pervade existing work. Cross-sectional designs cannot control for unobserved heterogeneity, the possibility that countries with high credit or trade exposure differ systematically in other dimensions affecting crisis vulnerability. Panel approaches with country fixed effects address this limitation by differencing out time-invariant characteristics, but most GFC studies do not employ such methods. Moreover, parallel trends assumptions receive limited attention. Whether high-exposure and low-exposure countries exhibited parallel growth

trajectories before the crisis, which would validate the difference-in-differences interpretation, is rarely tested explicitly through event studies or pre-trend analysis.

Third, heterogeneity across country types and structural characteristics remains underexplored.

Do credit and trade channels operate differently for developed versus developing countries?

Developed economies have deeper financial markets but also greater integration into global banking networks; developing economies may be more vulnerable to sudden stops in capital flows but less integrated into complex securitization chains. Existing studies provide limited answers to these questions, typically examining either narrow samples or pooled specifications that impose homogeneous effects across all countries.

III. Theoretical Framework

This paper is grounded in the theory of the financial accelerator mechanisms and international crisis transmission. Within this framework, countries are conceptualized as economies with varied levels of financial depth and trade integration. Before looking into the dominant mechanism between both credit and trade channels, it is necessary to establish the transmission mechanisms that they operate through.

At the microeconomic level, the financial accelerator of Bernanke and Gertler (1995) establishes that credit market imperfections amplify shocks beyond their initial magnitude. The forces at play between these imperfections arise from asymmetric information and moral hazard. When a negative shock reduces asset prices, the net worth for borrowers deteriorates. This causes lenders to demand higher risk premia or withdraw credit entirely, which leads to contractions in spending and investment that further depress asset prices and net worth. This leads to a self-reinforcing cycle that generates disproportionately large and persistent output losses relative to the size of the initial shock.

The financial accelerator operates through two distinct channels: the balance sheet channel and the bank lending channel. The balance sheet channel works through the borrower's side, whereby the fall in collateral values leads to firms and households cutting down on investment and consumption. On the other hand, the bank lending channel persists when the balance sheet of banks deteriorates through a rise in non-performing loans and falling asset values. This leads to a restriction upon the credit supply, amplifying the contractionary impulse beyond what the

balance sheet channel alone would generate. Both channels work together and drag each other downward.

The mechanisms that drive the financial accelerator operate through stock variables - the balance sheet channel and the leverage ratios. These are predetermined before a crisis hit, thus making the pre-crisis credit exposure an appropriate measurement. A country's vulnerability to accelerator dynamics was locked in before any agent could anticipate a shock. On the other hand, the trade channel operates through flow variables - the difference between export revenues and import expenditure. These adjust continuously and gradually as the crisis unfolds.

Building on the financial accelerator established by Bernanke and Gertler (1995), the key implication for this thesis' design is that pre-crisis credit depth acts as a proxy for how severely balance sheet and bank lending channels could bind when a shock hits. A country with a higher net domestic credit as a percentage of GDP would face a higher vulnerability for the accelerator dynamics relative to a country with a lower ratio, independently of how open it was to trade.

At the international level, Allen and Gale (2000) formalize how these balance sheet dynamics transmit across borders through overlapping claims in the banking sector. In normal times, cross-border interbank linkages serve as insurance against local liquidity shocks. However, these same buffers can become vectors of contagion when an unexpected aggregate shock occurs. To meet domestic withdrawal demands, banks liquidate their foreign holdings, which spreads financial distress through a chain of forced liquidations. What determines the severity of the contagion, however, lies within the structure of the financial network: incomplete networks,

where institutions are connected to only a limited number of counterparties, are most fragile because a shock to one node hits its neighbours with full force rather than being distributed across the system.

This logic is extended to cover exchange rate dynamics by Krugman (1999). Countries that carry substantial foreign-currency-denominated debt are at the risk of capital outflows triggered by financial crises. As they force depreciations within the currency, it forces an increase in the domestic value of their debt obligations; thereby, deteriorating the balance sheet of banks and firms and triggering further withdrawals in a self-reinforcing loop. As a result, financial shocks are transmitted across borders through lending and foreign-currency exposures, even in the complete absence of strong trade links between the affected economies. A country's vulnerability to this loop depends not on how much it trades with the crisis epicentre, but rather it becomes a case of concentrated integration. The major driver of vulnerability becomes large exposures channelled through a narrow set of financial linkages rather than either connectivity or depth in isolation.

The trade channel operates through an entirely different set of mechanisms, transmitting shocks gradually through the real economy rather than instantaneously through financial markets. The foundational mechanism is one of demand spillovers: a recession in one country reduces aggregate income, lowering demand for imports and transmitting negative demand shocks to trading partners through declining export revenues.

Beyond demand spillovers, the trade channel also operates through supply chain linkages. Given the interconnectedness of global production networks, intermediate goods cross borders multiple times before reaching final consumers. A contraction in the demand for those final goods, therefore, ripples backwards through the supply chain. This was shown by Levchenko, Lewis, and Tesar (2010), who documented that during the Global Financial Crisis, the fall in world trade far exceeded the decline in global GDP. This is because supply chain linkages amplified the demand shocks into a cascade of cross-border contractions.

Another mechanism is the terms-of-trade effect. Currency depreciations in crisis-hit countries shift the competitive landscape for trading partners. Countries that find themselves competing in the same export markets as the crisis epicenter may find their competitive position being eroded as depreciation of the currency makes exports of the crisis-country much cheaper. This terms-of-trade deterioration can reduce export revenues and dampen growth even in countries with no direct financial exposure to the crisis.

Despite the force of these mechanisms, the trade channel carries an important limitation compared to the credit channel - the speed of transmission. Obstfeld, Shambaugh, and Taylor (2010) note that the adjustment of the decrease in GDP is much slower for the trade channel as compared to the credit channel. Trade linkages transmit shocks gradually, giving economies time to adjust through currency depreciation, expenditure switching, and fiscal policy. Credit linkages propagate shocks almost instantaneously through financial markets, leaving little time for policy adjustment before balance sheet deterioration sets in.

Together, the theoretical framework predicts an asymmetric relationship in terms of the speed and the magnitude of transmission across both channels. The credit channel operates quicker, is self reinforcing, and acts independently of real linkages across economies. The trade channel, meanwhile, works more gradually and is bounded by the pace of real economy adjustment.

IV. Data

This paper draws on data from the cross-country panel dataset constructed by Lei, Li, and Mei (2023), which covers up to 180 countries over the period 2000 to 2019. It compiles information from several international databases such as the World Bank, the International Monetary Fund, and the UN National Accounts database. While the full dataset consists of 180 countries, the sample used within this thesis is reduced to 130 countries following listwise deletion of missing observations across key variables. The sample window from 2000 to 2019 is chosen deliberately: it spans the pre-crisis credit expansion of the early 2000s, the Global Financial Crisis, and a decade of recovery.

The primary measure of economic performance is real GDP at constant 2015 prices, sourced from the UN National Accounts database. GDP growth is computed as the log difference of real GDP across years, which approximates the annual percentage change in output.

Domestic credit exposure is measured using net domestic credit as a share of GDP, which is sourced from the World Bank Development Indicators. This variable is a broad measure of the banking system activity and encompasses both the private and the public sector, net of government deposits held at the banking system. In other words, net domestic credit is approximately equivalent to loans to the private sector plus loans to the government minus government deposits, and it is expressed as a share of GDP. This allows it to capture the depth of a country's financial system and its reliance on credit-based intermediation, which is amongst the most relevant dimensions of financial exposure for the transmission of crises. As shown with the

financial accelerator mechanism by Bernanke and Gertler (1995), countries with deeper credit markets are more likely to be exposed to fluctuations in asset prices and balance sheet deterioration.

The other main variable of interest is trade exposure, which is measured as the sum of exports and imports as a share of GDP, sourced from the World Bank World Development Indicators. This is used to measure the integratedness of a country's goods and services to the global economy. Within the dataset, the mean value of 0.794 and the median value of 0.809 reflect how trade openness is relatively uniformly distributed across countries as opposed to credit exposure. This contrast is imperative in the discussion of the two as potential transmission channels.

As for the control variables within the regression model, there are Government consumption as a share of GDP, FDI inflows as a share of GDP, mean years of schooling, and tradeable sector share of GDP.

Government consumption is added as a control for the fiscal policy stance of each country as it impacts GDP growth. FDI inflows capture capital account openness and foreign investment activity. To account for human capital accumulation, mean years of schooling, which is sourced from the UNDP Human Development Reports, acts as a proxy. This is in alignment with Mankiw, Romer, and Weil (1992). Finally, the tradeable sector share of GDP looks at the contribution of manufacturing, mining, and other industries to control for sectoral composition, which may impact growth outcome as well as crisis transmission patterns.

Lastly, country-specific financial crises episodes are identified using the datasets of Hardy and Sever (2021) and Laeven and Valencia (2020), which together provide comprehensive coverage of banking crises, sovereign debt crises, and currency crises across the full sample period. The crisis variable is not restricted to the Global Financial Crisis, as it captures all crisis episodes. The crisis variable covers country-specific banking crises, periods of sovereign stress, and currency collapses. For the main regression specification and the probit model, the GFC is defined as the period from 2007 to 2009 and the crisis dummy variable used falls between this period.

By focusing on the pre-crisis variation across countries, this paper looks at exposure to credit and trade channels to identify the dominant mechanism of GFC transmission. Since during the Global Financial Crisis, a country's credit exposure and trade openness was highly correlated with its growth outcome - the transmission channels are fixed at their pre-crisis values, specifically to 2006. In the case that data from 2006 was not available, values from 2005 are used as a fallback. This pre-determination ensures that the exposure measures are not contaminated by the crisis itself, since they are measured before the shock occurred. This approach follows the identification strategy by Kalemli-Ozcan et al. (2013). The intuition is straightforward: countries with deeper credit market exposures in 2006 - before agents could anticipate the severity of the shock - were differently exposed to the crisis. The same holds true for trade openness. Thus, by interacting these variables with the GFC period indicator, the specification is able to identify the causal impact of each channel on growth outcomes.

The following is a table describing the different variables used in the empirical analysis throughout this paper.

Table 1: Summary Statistics — Full Sample (2000–2019)

Variable	Obs	Mean	Std. Dev.	Median	Min	Max
<i>Panel A: Dependent Variable</i>						
GDP Growth	3,439	0.035	0.053	0.036	−0.698	0.671
<i>Panel B: Pre-Crisis Exposure Variables (2006)</i>						
Pre-Crisis Credit Exposure	3,080	0.388	0.399	0.245	−0.394	1.495
Pre-Crisis Trade Exposure	2,860	0.817	0.310	0.809	0.260	1.646
<i>Panel C: Control Variables</i>						
Government Consumption (% GDP)	2,887	0.166	0.070	0.162	0.010	0.741
FDI Inflows (% GDP)	3,196	0.045	0.061	0.030	−0.283	0.474
Mean Years of Schooling	3,269	8.161	3.109	8.382	1.100	14.152
Trade Sector Share of GDP	3,600	0.311	0.142	0.297	0.020	0.825
<i>Panel D: Crisis Indicators</i>						
Financial Crisis Dummy	2,840	0.060	0.237	0	0	1
GFC Dummy (2007–2009)	3,620	0.150	0.357	0	0	1
Developed Economy Dummy	3,620	0.155	0.362	0	0	1

Notes: The sample covers up to 181 countries from 2000 to 2019, reducing to approximately 130 countries in the regression sample after listwise deletion of missing observations. GDP Growth is the log difference of real GDP at constant 2015 prices from the UN National Accounts database. Pre-Crisis Credit Exposure is net domestic credit as a share of GDP fixed at 2006 values (or 2005 if 2006 is missing). Pre-Crisis Trade Exposure is the sum of exports and imports as a share of GDP, fixed at 2006 values. Developed Economy classification follows the IMF Advanced Economies list. Financial crisis data are from Hardy and Sever (2021) and Laeven and Valencia (2020).

Table 1 presents the summary statistics for these variables across the full sample of 181 countries over 2000 to 2019. The four panels within the table cover the dependent variable, the pre-crisis exposure variables, the control variables, and the crisis indicators.

The average annual GDP growth rate across the sample is 3.5 percent with a standard deviation of 5.3 percent. The extreme values correspond to documented real-world events, which include

sharp economic contractions during periods of conflict (e.g., Libya 2011, Iraq 2003, Yemen 2015, Central African Republic 2013) and subsequent post-conflict recoveries.

As mentioned earlier, the pre-crisis variables are fixed to 2006, and thus they remain constant in the table. Pre-crisis credit exposure averages 38.8 percent of GDP across the full sample, though the distribution is highly right-skewed. On the other hand, pre-crisis trade exposure averages 81.7 percent of GDP with a median of 80.9 percent.

Among the control variables, government consumption averages 16.6 percent of GDP with a median of 16.2 percent. FDI inflows average 4.5 percent of GDP with a median of 3.0 percent, and the distribution is right-skewed. The minimum values that are negative observations reflect years in which disinvestment or profit repatriation exceeded new inflows, which is a legitimate economic phenomenon particularly during crisis years. Mean years of schooling averages 8.2 years. The tradable sector share of GDP averages 31.1 percent, which indicates that roughly one-third of domestic value added is generated in sectors such as agriculture, mining, and manufacturing. This measure reflects the composition of domestic production and is distinct from trade exposure, which captures total trade flows relative to GDP.

The dummy for financial crises has a mean of 0.060, which indicates that approximately 6% of observations across countries are classified as crisis episodes. When restricted to the GFC dummy, this increases to a mean value of 0.15.

To examine how macroeconomic conditions varied across countries, it is necessary to decompose the sample into different periods. Table 2 further disaggregates the summary statistics across 3 periods: pre-crisis (2000-2006), GFC (2007-2009), and post-crisis (2010-2019). This is shown in table 2.

Table 2: Summary Statistics by Crisis Period

Variable	Pre-Crisis (2000–2006)		GFC (2007–2009)		Post-Crisis (2010–2019)	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
<i>Panel A: Dependent Variable</i>						
GDP Growth	0.043	0.050	0.030	0.064	0.031	0.050
<i>Panel B: Pre-Crisis Exposure Variables</i>						
Pre-Crisis Credit Exposure	0.388	0.399	0.388	0.399	0.388	0.399
Pre-Crisis Trade Exposure	0.817	0.310	0.817	0.310	0.817	0.310
<i>Panel C: Control Variables</i>						
Government Consumption (% GDP)	0.161	0.063	0.162	0.073	0.170	0.074
FDI Inflows (% GDP)	0.041	0.054	0.058	0.067	0.043	0.063
Mean Years of Schooling	7.508	3.057	8.042	3.116	8.636	3.060
<i>Panel D: Observations</i>						
N (GDP Growth)	1,086		543		1,810	

Notes: Crisis periods are defined as follows: Pre-Crisis (2000–2006), Global Financial Crisis (2007–2009), and Post-Crisis (2010–2019). Pre-crisis exposure variables are fixed at 2006 values and are therefore identical across periods by construction. GDP growth declined from an average of 4.3% in the pre-crisis period to 3.0% during the GFC, with cross-country dispersion increasing substantially (standard deviation rising from 5.0% to 6.4%), indicating significant divergence in outcomes across countries during the crisis. Post-crisis growth recovered only partially to 3.1%, remaining below pre-crisis levels, consistent with the scarring effects documented in the literature. Mean years of schooling increases across periods reflecting global improvements in educational attainment over the sample period.

An important revelation from this table is the way in which average global GDP growth fell from 4.3 percent prior to the crisis to 3 percent during the GFC. The standard deviation grew from 5 percent to 6.4 percent, which reflects the widening distribution across the GFC. This showcases the heterogenous transmission of the crisis, where all countries were affected in different ways by the crisis. The fact that dispersion increased substantially suggests that differential exposure to transmission channels played a meaningful role in determining which countries fared better or worse during the shock.

Post-crisis growth recovered only partially to 3.1 percent over 2010 to 2019, remaining persistently and significantly below the pre-crisis average of 4.3 percent. This is consistent with the literature, where Cerra and Saxena (2008) document that financial crises are associated with permanent output level losses, with no tendency for post-crisis catch-up growth. Moreover, Reinhart and Rogoff (2009) similarly find that recovery from financial crises is characterised by slow and incomplete rebound, and that output per capita often fails to return to its pre-crisis trend within a decade.

Trade openness remained mostly stable across periods in ratio terms, consistent with the findings by Levchenko, Lewis, and Tesar (2010), which states that despite collapsing trade volumes in absolute terms, trade share of GDP did not change as dramatically because of contractions within GDP.

Government consumption remained relatively stable through the pre- and crisis periods between 16.1 and 16.2 percent, and finally rose to 17 percent post-crisis. This increase is consistent with the government's role as a stabiliser in the fiscal policy literature.

A surprising finding is the value of FDI inflows, which were higher during the GFC as opposed to the pre-crisis period.

To further examine the heterogeneity of the crisis, table 3 looks at the summary statistics of countries based on development status. Country development status is classified using the IMF Advanced Economies list, which designates 37 countries as advanced economies. Due to missing data across one or more variables, the subsample for the developed country analysis falls to 22

countries, while the developing country subsample includes up to 109 countries. This is shown in table 3.

Table 3: Summary Statistics by Development Status

Variable	Developing Countries			Developed Countries		
	N	Mean	Std. Dev.	N	Mean	Std. Dev.
GDP Growth	2,907	0.037	0.056	532	0.020	0.027
Pre-Crisis Credit Exposure	2,580	0.256	0.261	500	1.069	0.272
Pre-Crisis Trade Exposure	2,360	0.819	0.305	500	0.806	0.331
Government Consumption (% GDP)	2,327	0.159	0.074	560	0.195	0.039

Notes: Country classification follows the IMF Advanced Economies list. Developed countries include up to 37 economies, reducing to 22 in the regression sample. Pre-crisis credit exposure is substantially higher in developed economies (106.9% of GDP) compared to developing economies (25.6% of GDP), reflecting deeper financial markets in advanced economies. By contrast, pre-crisis trade exposure is virtually identical across groups (81.9% vs 80.6% of GDP), providing useful variation for identifying the credit channel separately from the trade channel.

Pre-crisis credit exposure averaged 106.9 percent of GDP among developed economies, which means that, on average, net domestic credit outstanding exceeded the entire annual output of the economy in advanced countries in 2006. On the other hand, among developing economies, the corresponding figure was just 25.6 percent. This four-fold difference reflects the fundamentally different structure of financial systems across development groups. Developed economies were categorized by large and sophisticated banking networks, developed corporate bond and mortgage markets, and a broad range of non-banking financial institutions. By contrast, developing economies are characterized by having shallower banking systems with lower financial inclusion and a greater reliance on informal credit markets.

As for trade exposure across developed and developing economies, it is economically negligible with a difference of a meagre 1.6 percentage points.

The other variables between development groups show expected patterns. Developed economies had higher government consumption - 19.5 percent compared to 15.9 percent - reflecting the public service provision characteristic of developed economies. GDP growth was substantially lower for developed economies at 2 percent per year than for developing economies at 3.7 percent per year, which is consistent with the conditional convergence theory - poorer countries grow faster than richer ones when starting from a lower capital-to-labour ratio. This, however, is accompanied by substantially higher volatility: the standard deviation of GDP growth was 5.6 percent for developing economies versus just 2.7 percent for developed economies.

V. Empirical Strategy and Results

Building on the theoretical framework, the following hypotheses emerge:

H₁: Countries with greater pre-crisis credit exposure experienced systematically larger output contractions during the Global Financial Crisis.

H₂: The dominance of the credit channel over the trade channel is stable across both crisis incidence and intensity outcomes.

To estimate the impact of transmission of crises, this thesis adopts a range of estimation techniques, each carrying distinct identifying assumptions, in order to assess the sensitivity of the credit and trade channels. The estimation frameworks include: a country fixed effects panel regression as the baseline specification, a two-way fixed effects model, a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), and a probit model for crisis probability.

The baseline specification is a pooled OLS regression of the form:

$$\Delta \ln \text{GDP}_{it} = \alpha + \beta_1(\text{GFC}_t \times \text{CreditExposure}_{i, 2006}) + \beta_2(\text{GFC}_t \times \text{TradeExposure}_{i, 2006}) + X'_{it}\gamma + \varepsilon_{it}$$

where α denotes the intercept term ; X'_{it} is a vector of time-varying control variables such as government consumption, FDI inflows, mean years of schooling, and tradeable sector share of

GDP; and ε_{it} is the error term, clustered at the country level. The coefficients β_1 and β_2 are the primary variables of interest, which reflect the average effect of pre-crisis credit and trade exposure onto GDP growth during the GFC across the full sample.

The GFC indicator is a binary variable that takes the value of one for all countries during 2007, 2008, and 2009, and zero otherwise. β_1 and β_2 therefore identify how countries with differing levels of pre-crisis exposure fared during the crisis years relative to their own non-crisis growth performance, pooling all variation across countries and time periods without controlling for time-invariant country characteristics.

This is where the interaction terms come into play. Rather than estimating the average effect of the GFC across all countries, β_1 and β_2 identify the differential effect: the additional reduction in growth experienced by countries with greater pre-crisis exposure relative to those with lower exposure. The common global shock of the GFC is not separately absorbed; its effect on growth is captured through these exposure interactions, which means that the coefficients reflect how the severity of the crisis covaried with pre-crisis financial and trade conditions.

Table 4 presents the results of the baseline specification. Column 1 and 2 estimate the credit and trade channel in isolation, column 3 is the horse-race specification, column 4 and 5 further add controls such as FDI inflows, mean years of schooling, and the tradeable sector share of GDP. All five specifications cluster standard errors at the country level.

Table 4: Crisis Transmission Channels and GDP Growth - Pooled OLS

	(1) Credit Channel	(2) Trade Channel	(3) Horse Race	(4) Extended	(5) Full Controls
<i>Panel A: Key Variables of Interest</i>					
GFC $\times$ Credit Exposure	-0.0169*** (0.0041)		-0.0176*** (0.0063)	-0.0160*** (0.0060)	-0.0123* (0.0065)
GFC $\times$ Trade Exposure		-0.0070** (0.0033)	0.0017 (0.0046)	-0.0011 (0.0046)	-0.0029 (0.0048)
<i>Panel B: Control Variables</i>					
Government Consumption	-0.2341*** (0.0367)	-0.1559*** (0.0410)	-0.2456*** (0.0382)	-0.2272*** (0.0469)	-0.2201*** (0.0474)
FDI Inflows				0.1047*** (0.0304)	0.1088*** (0.0308)
Mean Years of Schooling				-0.0012** (0.0005)	-0.0002 (0.0006)
Trade Sector Share					0.0402** (0.0193)
<i>Panel C: Model Diagnostics</i>					
Country FE	No	No	No	No	No
Year FE	No	No	No	No	No
Observations	2,567	2,595	2,435	2,367	2,367
Countries	141	140	131	130	130
R^2	0.076	0.053	0.079	0.098	0.104

Notes: All models estimated using pooled OLS with standard errors clustered at the country level. Robust standard errors reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The results from the Pooled OLS regression offer a transparent view of the data, as they do not account for fixed effects or absorbed shocks. Rather, they measure the raw relationship between pre-crisis exposure and GDP outcomes across the sample. The horse-race specification in column 3 places both channels simultaneously, which makes it the natural starting point for the discussion. As seen, the pre-crisis credit exposure has a coefficient of -0.0176, which is significant at the 1 percent level. On the other hand, trade exposure remains negligible with the value of 0.0017, while not being statistically significant. In this raw specification without any added controls, the credit channel dominates.

Looking at the channels in isolation portrays a similar story. The trade exposure, in column 2, has a coefficient of -0.0070, which is significant at the 5 percent level. While on its own, it can be deemed to have a transmission role for crises; however, once the credit channel enters - the coefficient flips its sign and loses its statistical significance.

As for the controls, government consumption is consistently negative and strongly significant across all five specifications, with coefficients ranging from -0.156 to -0.246. This, however, could be a reflection of reverse causality, wherein governments expand their fiscal policy during economic downturns. FDI inflows are positive and significant at the 1 percent level once included. Mean years of schooling is negative and significant in column 4 but loses significance in column 5. Finally, the tradeable sector share of GDP is positive and significant at the 5 percent level in column 5 with a coefficient of 0.0402.

While it presents a raw version of the results, it rests on the basic assumption: neither country nor year fixed effects play a role. Thus, coefficients in the table reflect not just within-country dynamics, but also systemic differences across advanced and developing economies. Thus, the natural next step becomes to introduce country fixed effects, which differentiate these time-invariant country characteristics and isolate the within-country variation that more credibly identifies the transmission channel.

Introducing country fixed effects helps with the tightening of the identification. Where the pooled OLS pooled all variation across countries and time, the country fixed effects specification differences out time-invariant country characteristics. This accounts for geographic variation, institutional history, and long-run growth trends, which leaves only within-country variation to

identify the channel effects. The question shifts from whether countries with high-credit or trade exposure grew more slowly than their low exposure counterparts, but whether the same country grew more slowly during the GFC relative to its non-crisis performance.

The specification of the country Fixed Effects regression thus takes the following form:

$$\Delta \ln \text{GDP}_{it} = \alpha_i + \beta_1(\text{GFC}_t \times \text{CreditExposure}_{i, 2006}) + \beta_2(\text{GFC}_t \times \text{TradeExposure}_{i, 2006}) + \mathbf{X}'_{it}\gamma + \varepsilon_{it}$$

where α_i denotes country fixed effects that absorb all time-invariant country characteristics; $\mathbf{X}'_{it}$ is a vector of time-varying control variables; and ε_{it} is the error term, which is clustered at the country level. The coefficients β_1 and β_2 are the primary variables of interest, as they look at the differential effect of credit and trade exposure on GDP growth during the Global Financial Crisis.

The results for this specification are shown in table 5.

Table 5: Crisis Transmission Channels and GDP Growth - Country Fixed Effects

	(1) Credit Channel	(2) Trade Channel	(3) Horse Race	(4) Extended	(5) Full Controls
<i>Panel A: Key Variables of Interest</i>					
GFC $\times$ Credit Exposure	-0.0155*** (0.0042)		-0.0119* (0.0069)	-0.0125* (0.0070)	-0.0123* (0.0070)
GFC $\times$ Trade Exposure		-0.0096*** (0.0033)	-0.0033 (0.0051)	-0.0053 (0.0052)	-0.0052 (0.0051)
<i>Panel B: Control Variables</i>					
Government Consumption	-0.4131*** (0.0991)	-0.4227*** (0.1012)	-0.4262*** (0.1038)	-0.4034*** (0.1109)	-0.3821*** (0.0987)
FDI Inflows				0.1114*** (0.0318)	0.1156*** (0.0333)
Mean Years of Schooling				-0.0038** (0.0017)	-0.0022 (0.0024)
Trade Sector Share					0.0650 (0.0576)
<i>Panel C: Model Diagnostics</i>					
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	No	No	No
Observations	2,567	2,595	2,435	2,367	2,367
Countries	141	140	131	130	130
Within R^2	0.054	0.053	0.055	0.067	0.069

Notes: All models estimated using one-way fixed effects (country only) with standard errors clustered at the country level, reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

As seen in the table, even after the tightening, the results corroborate the pooled OLS results.

Within the horse-race specification of column 3, the credit channel coefficient stands at -0.0119, statistically significant at the 10 percent level, which is a modest shrinkage from the pooled OLS estimate. The shrinkage reflects that some of what the pooled OLS attributed to credit transmission was, in fact, capturing systematic differences between high and low credit countries unrelated to the GFC. The fate of the trade channel, however, does not change. It remains negligible and statistically insignificant, mirroring the pooled OLS findings.

What remains striking is the behaviour of the trade channel in isolation. The coefficient for this is -0.0096, which is statistically significant at the 1 percent level. This is strengthened when placed

side by side with the pooled OLS, where it was significant at the 5 percent level. However, similar to what was seen in the OLS specification, once credit enters the playing field - the trade exposure variable loses all significance.

Among the controls, the fixed effects specification reveals one important departure from the pooled OLS. The coefficients for government consumption lie between -0.382 and -0.426, which are substantially larger than the results in the pooled OLS. FDI inflows and mean years of schooling follow the same pattern as before - positive and statistically significant, and negative, then losing significance.

The country fixed effects specification therefore corroborates the findings of the pooled OLS regressions: the credit channel survives, whereas the trade channel does not. The question then remains whether the common global shock of the GFC is contaminating the interaction terms. That is where the two-way fixed effects specification comes into play, as shown in table 6.

Despite the findings of the country fixed effects, one concern remains: the GFC period indicator entered entirely through the interaction terms, meaning the common global shock hitting all economies simultaneously in 2007-2009 was not separately absorbed. To address this, the TWFE model adds year fixed effects alongside country fixed effects to absorb all common global variation. This allows our coefficients of interest β_1 and β_2 to identify only what remains after both country heterogeneity and the global shock are removed: why some countries fared worse than others during the same crisis years.

The specification, therefore, has a difference-in-differences structure, with two comparisons being made simultaneously. The first comparison is within each country over time: how did a country's growth rate during the GFC compare to its own growth performance outside the crisis window. The second comparison is across countries during the crisis: how did countries with higher exposure to credit and trade channels fare systematically compared to low-exposure countries during the GFC years. Since the GFC indicator is now fully absorbed by the year fixed effects for 2007, 2008, and 2009, β_1 and β_2 identify only the differential effect experienced by countries based on their pre-crisis exposure, above and beyond the average global contraction.

Table 6: Crisis Transmission Channels and GDP Growth - TWFE

	(1) Credit Channel	(2) Trade Channel	(3) Horse Race	(4) Extended	(5) Full Controls
<i>Panel A: Key Variables of Interest</i>					
GFC $\times$ Credit Exposure	-0.0121* (0.0065)		-0.0118* (0.0069)	-0.0120 (0.0072)	-0.0119 (0.0072)
GFC $\times$ Trade Exposure		-0.0034 (0.0084)	0.0014 (0.0086)	0.0022 (0.0087)	0.0021 (0.0087)
<i>Panel B: Control Variables</i>					
Government Consumption	-0.3413*** (0.1095)	-0.3423*** (0.1104)	-0.3448*** (0.1148)	-0.3327*** (0.1240)	-0.3254*** (0.1130)
FDI Inflows				0.0871*** (0.0301)	0.0887*** (0.0315)
Mean Years of Schooling				0.0028 (0.0038)	0.0028 (0.0038)
Trade Sector Share					0.0276 (0.0664)
<i>Panel C: Model Diagnostics</i>					
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	2,567	2,595	2,435	2,367	2,367
Countries	141	140	131	130	130
Within R^2	0.112	0.112	0.109	0.115	0.115

Notes: All models estimated using two-way fixed effects (country and year) with standard errors clustered at the country level, reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The central finding of this paper is established in column 1: the credit channel. The coefficient on $GFC_t \times CreditExposure_{i, 2006}$ is -0.0121 and statistically significant at the 10 percent level. The economic magnitude of this effect is substantial. A one unit increase in pre-crisis credit exposure is associated with a 1.21 percentage point reduction in annual GDP growth during the GFC period. In terms of standard deviation, scaled to one standard deviation of credit exposure in the sample (0.40 units), the implied growth reduction is approximately 0.48 percentage points per year. For reference, the average GDP growth decline from the pre-crisis to the GFC period was 1.3 percentage points. This is a large effect and is consistent with the emphasis placed on the credit channel in the theoretical and empirical literature on the GFC.

The credit channel coefficient is stable across all five specifications. Moving from the bivariate credit channel specification to the horse-race specification that controls for trade exposure changes the coefficient by just 0.0003, confirming that the credit channel result is not confounded by omitted trade exposure. In column 4 and 5, as more controls are added, the change is negligible at -0.0119. However, the credit channel coefficient loses marginal statistical significance in columns 4 and 5. This loss of significance does not reflect a change in the point estimate - as established above, the coefficient is essentially unchanged - but rather reflects a modest increase in the standard error associated with the reduction in sample size from 141 countries in column 1 to 130 countries in columns 4 and 5.

As for the trade channel, the results are consistently insignificant across all specifications. In column 2, the coefficient on $GFC_t \times TradeExposure_{i, 2006}$ is -0.0034. Although the point estimate

suggests a small negative effect, the confidence interval is wide - including both positive and negative values. In the horse-race specification (column 3), the coefficient reverses sign to +0.0014 and remains highly insignificant, and this pattern persists in columns 4 and 5, where the estimates remain close to zero.

Although global trade volumes declined sharply during the crisis, this aggregate shock is absorbed by year fixed effects. The results instead capture whether more open economies experienced relatively worse outcomes, and the consistently null findings suggest they did not. This implies that the trade collapse was broadly shared across countries, without generating differential growth effects based on pre-crisis trade openness.

As for the control variables, government consumption is consistently negative and highly significant across all five specifications, with coefficients ranging from -0.325 to -0.345. The interpretation of this coefficient in a two-way fixed effects specification is that within a given country, years with higher government consumption as a share of GDP are associated with lower GDP growth. This negative within-country relationship could reflect several mechanisms: reverse causality, whereby governments increase spending during downturns; genuine crowding out of private investment; or fiscal sustainability concerns associated with high government spending that reduce private sector confidence.

FDI inflows are positive and significant at the 1 percent level in columns 4 and 5 (coefficients of 0.087 and 0.089, respectively). The positive effect of FDI on growth aligns with the idea that

foreign direct investment supplements domestic savings, facilitates technology and knowledge transfer, and helps integrate economies into global value chains.

A drawback of using the two-way fixed effects specification is that it assumes that the treatment effect of the GFC is homogenous across countries and all time periods. This assumption is restrictive as countries entered into financial crises at different points during the GFC window, with the United States experiencing the crisis from 2007 and many European countries entering the crisis much later. As shown by Goodman-Bacon (2019) and de Chaisemartin and D'Haultfoeuille (2020), this can potentially bias the overall estimates as TWFE in staggered adoption settings can assign negative weights to some group-time average treatment effects. To assess the sensitivity of the main findings to this assumption, the difference-in-differences estimator of Callaway and Sant'Anna (2021) is applied as a robustness check.

The CS DiD approach differs from the TWFE specification in two important ways. Firstly, rather than using the global GFC dummy applied uniformly to all countries, treatment is defined as entry into a formal financial crisis as classified by Hardy and Sever (2021) and Laeven and Valencia (2020). While countries such as China and India did not experience formal crises and therefore form part of the control group, their role should be interpreted with caution. The absence of crises in these economies may reflect structural differences such as financial regulation, capital account openness, and policy responses that distinguish them from crisis-affected countries. Therefore, they may not represent perfect counterfactuals. However, within this framework, identification does not rely on any single country acting as a control. Instead, the CS DiD framework constructs comparison groups using all not-yet-treated and

never-treated countries, and identification rests on a conditional parallel trends assumption across these groups. This framework allows for heterogeneity in levels and structural characteristics across countries, provided that such differences do not generate differential trends in the absence of treatment.

Second, rather than estimating a single average treatment effect, the CS DiD estimator identifies group-time average treatment effects of the form:

$$ATT(g,t) = E[Y_t(g) - Y_t(0) \mid G_g = 1]$$

where $Y_t(g)$ denotes the potential GDP growth outcome at time t for a country first entering a crisis in period g . $Y_t(0)$ denotes the counterfactual outcome in the absence of crisis entry, and $G_g = 1$ indicates membership in cohort g , which is defined as the year a country first entered a formal financial crisis. These group-time average treatment effects are then aggregated into an overall average treatment effect on the treated:

$$\theta^0 = \sum_{g \in G} \sum_{t \geq g} w(g,t) \cdot ATT(g,t)$$

where $w(g,t)$ are weights proportional to cohort size and G denotes the set of treated cohorts. In this case, the identification relies on a conditional parallel trends assumption: in the absence of treatment, the average outcomes for cohort g and the not-yet-treated comparison group would have followed parallel paths conditional on observed covariates. The not-yet-treated comparison group is preferred over the never-treated group following Assumption 5 of Callaway and

Sant'Anna (2021), as it allows more countries to serve as valid controls and improves precision given the sample size.

Within this framework, the treatment is binary as it depends on formal crisis entry, while pre-crisis credit and trade exposure act as pre-determined country characteristics, which are used to examine heterogeneity in treatment effects. Therefore, the CS DiD estimates identify the causal effect of crisis entry. Differences across countries based on their exposure to credit and trade reflect how pre-crisis conditions amplify or mitigate this effect rather than constituting treatment intensity in the DiD sense.

The construction of the sample is as follows: countries with pre-GFC crisis episodes are excluded to ensure a clean treatment and control group, as these countries cannot serve as credible counterfactuals. The cohort variable is defined as the first year each country entered a formal financial crisis during 2007 to 2009. Due to a small sample size of countries entering the 2007 and 2009 cohorts, they collapsed into the 2008 cohort. This yields a clean two-group design: 31 treated countries that entered crisis during 2007 to 2009 and 75 never-crisis control countries. Two sample windows are estimated throughout: a short window of 2000 to 2009 to maintain direct comparability with the TWFE baseline, and a full window of 2000 to 2019 to examine whether findings persist over the post-crisis recovery decade.

The baseline results for the CS DiD methodology are presented in Table 8. The overall ATT is negative 0.0037 in the full window and negative 0.0111 in the short window, which are both statistically insignificant. This is expected, as the baseline ATT averages the effect of crisis entry

across all treated countries, abstracting from differences in pre-crisis exposure. As a result, it masks substantial heterogeneity in treatment effects that emerges once countries are grouped by credit and trade exposure. The more informative results come from the heterogeneity analysis, which are presented in table 7.

Table 7 examines heterogeneity in treatment effects by pre-crisis credit and trade exposure. For the credit channel, the results align with the TWFE findings. The estimated effect of crisis entry is negative for high-credit countries, with short-run ATT estimates of approximately negative 0.011, while low-credit countries exhibit effects close to zero. This supports the idea that deeper financial systems amplified the initial shock. At the same time, high-credit countries show stronger recovery over time, with positive and increasing ATT estimates after 2015.

Table 7: Staggered Difference-in-Differences: Baseline Results

	Coefficient	Std. Error	z	p-value
<i>Panel A: Pre-Treatment Periods (Parallel Trends Test)</i>				
Tm_6 (2001–02)	−0.0096**	(0.0043)	−2.24	0.025
Tm_5 (2002–03)	−0.0059	(0.0054)	−1.11	0.268
Tm_4 (2003–04)	0.0067	(0.0063)	1.05	0.293
Tm_3 (2004–05)	−0.0061	(0.0053)	−1.16	0.245
Tm_2 (2005–06)	0.0005	(0.0046)	0.10	0.921
Tm_1 (2006–07)	−0.0045	(0.0052)	−0.86	0.390
<i>Panel B: Post-Treatment Periods</i>				
Tp_0 (2007–08)	−0.0070	(0.0097)	−0.72	0.473
Tp_1 (2007–09)	−0.0151	(0.0183)	−0.83	0.408
Tp_2 (2007–10)	−0.0037	(0.0109)	−0.34	0.730
Tp_3 (2007–11)	−0.0076	(0.0096)	−0.79	0.428
Tp_4 (2007–12)	−0.0060	(0.0092)	−0.65	0.518
Tp_5 (2007–13)	−0.0030	(0.0089)	−0.34	0.733
Tp_6 (2007–14)	0.0000	(0.0100)	0.00	0.999
Tp_7 (2007–15)	−0.0055	(0.0111)	−0.50	0.619
Tp_8 (2007–16)	−0.0030	(0.0085)	−0.35	0.725
Tp_9 (2007–17)	0.0002	(0.0076)	0.02	0.984
Tp_{10} (2007–18)	0.0053	(0.0079)	0.67	0.503
Tp_{11} (2007–19)	0.0015	(0.0077)	0.20	0.845
<i>Panel C: Aggregated Effects</i>				
Pre-treatment average	−0.0032**	(0.0015)	−2.16	0.031
Post-treatment average	−0.0037	(0.0079)	−0.47	0.642
Overall ATT	−0.0037	(0.0079)	−0.46	0.643
<i>Panel D: Model Diagnostics</i>				
Observations		1,892		
Treated countries		31		
Control countries		75		
Pre-trends $\chi^2(6)$		10.886		
Pre-trends p-value		0.092		

Notes: Estimates are group-time average treatment effects $ATT(g, t)$ from Callaway and Sant'Anna (2021). The treatment group consists of 31 countries that first entered formal financial crisis during 2007–2009 as classified by Hardy and Sever (2021) and Laeven and Valencia (2020), collapsed into a single 2008 cohort. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: CS DiD: Credit and Trade Channel Transmission

	(1) High Credit	(2) Low Credit	(3) High Trade	(4) Low Trade
<i>Panel A: Overall ATT</i>				
ATT	0.0004 (0.0062)	−0.0036 (0.0186)	0.0206* (0.0122)	−0.0114 (0.0102)
<i>Panel B: Post-Treatment Group-Time Effects</i>				
Tp_0 (2007–08)	−0.0107 (0.0105)	0.0030 (0.0116)	0.0200 (0.0144)	−0.0092 (0.0071)
Tp_1 (2007–09)	−0.0120 (0.0148)	−0.0001 (0.0423)	0.0366 (0.0293)	−0.0349** (0.0175)
Tp_4 (2007–12)	−0.0105 (0.0101)	−0.0013 (0.0225)	0.0199 (0.0135)	−0.0239 (0.0158)
Tp_7 (2007–15)	0.0127* (0.0066)	−0.0466* (0.0264)	0.0109 (0.0180)	−0.0041 (0.0147)
Tp_8 (2007–16)	0.0066 (0.0048)	−0.0261 (0.0239)	0.0111 (0.0146)	−0.0026 (0.0123)
Tp_{10} (2007–18)	0.0115 (0.0071)	0.0044 (0.0148)	0.0272** (0.0129)	−0.0050 (0.0082)
Tp_{11} (2007–19)	0.0084 (0.0068)	−0.0054 (0.0154)	0.0203* (0.0118)	0.0001 (0.0073)
<i>Panel C: Model Diagnostics</i>				
Observations	989	836	851	978
Comparison group		Not yet treated		
Sample period		2000–2019		

Notes: Each column reports ATT estimates from separate Callaway and Sant’Anna (2021) estimations on subsamples split at the sample median. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

For the trade channel, the heterogeneity is even more pronounced. The estimated effect of crisis entry is consistently less negative, and in some periods positive, for high-trade countries, with the overall ATT reaching 0.0206. This becomes marginally significant at the 10 percent level in the full window. In contrast, low-trade countries experience more negative effects, including a significant short-run decline. This pattern suggests that trade openness mitigates the impact of crisis exposure, rather than acting as a transmission channel.

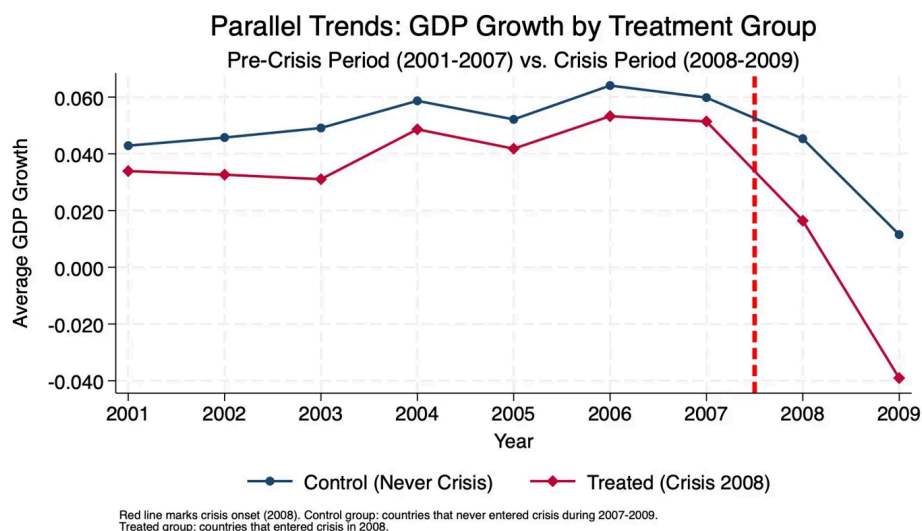
Table 8 compares the estimates of ATT across the two sample windows - the short window (2000-2009) and the long window (2000-2019). The overall ATT falls from -0.0111 in the short window to -0.0037 in the full window, which reflects gradual post-crisis recovery among treated countries.

With regards to the credit channel, the difference between high credit and low credit exposure ATTs is negative 0.0128 in the short window and 0.0040 in the full window. This reflects the stronger initial impact on high credit countries, followed by their faster long-run recovery. As for the trade channel, the difference is positive and significant in both windows - 0.0503 in the short window and 0.0320 in the full window. This reaffirms the role of trade openness as a layer of protection.

Table 9: CS DiD: Short vs. Long Window Comparison

	2000–2009	2000–2019
<i>Panel A: Overall ATT</i>		
Overall ATT	−0.0111 (0.0128)	−0.0037 (0.0079)
<i>Panel B: Credit Channel (ATT by Subsample)</i>		
High credit ATT	−0.0114 (0.0109)	0.0004 (0.0062)
Low credit ATT	0.0014 (0.0246)	−0.0036 (0.0186)
Difference (High – Low)	−0.0128	0.0040
<i>Panel C: Trade Channel (ATT by Subsample)</i>		
High trade ATT	0.0283 (0.0201)	0.0206* (0.0122)
Low trade ATT	−0.0220** (0.0091)	−0.0114 (0.0102)
Difference (High – Low)	0.0503**	0.0320*
<i>Panel D: Model Diagnostics</i>		
Observations	904	1,892
Treated countries	31	31
Control countries	75	75
Pre-trends $\chi^2(6)$	10.886	10.886
Pre-trends p-value	0.092	0.092

Notes: This table compares key ATT estimates across two sample windows. The 2000–2009 window maintains direct comparability with the TWFE baseline. The 2000–2019 window captures long-run post-crisis dynamics. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 2. Parallel Trends Assumption

Notes: Figure 2 plots the GDP growth trajectories for treated and control countries. During the pre-crisis period, both groups exhibit similar growth dynamics, with control countries averaging 5.3% annual growth and treated countries averaging 4.4%. Following the onset of the crisis in 2008, the trajectories diverged sharply: control countries experienced a moderate decline to 1.2% growth by 2009, while treated countries contracted by 4.0%. This pattern supports the validity of the difference-in-differences identification strategy.

The three estimation frameworks each rest on different identifying assumptions, making their comparison a meaningful robustness test rather than a formality. The country FE baseline controls for time-invariant heterogeneity but leaves common global shocks unabsorbed; TWFE additionally removes year-level variation; and CS DiD abandons the uniform GFC dummy entirely, defining treatment as formal crisis entry against a not-yet-treated comparison group. If the credit channel finding were an artifact of any particular modelling choice, it would be expected to break down as these assumptions change. Table 9 shows that it does not.

Table 10: Robustness Summary: Pooled OLS vs. Country FE vs. TWFE vs. CS DiD

	Pooled OLS		Country FE		TWFE		CS DiD	
	Credit Channel	Trade Channel	Credit Channel	Trade Channel	Credit Channel	Trade Channel	Credit Channel	Trade Channel
Panel A: Full Window Estimates (2000–2019)								
High exposure estimate	−0.0176*** (0.0063)	0.0017 (0.0046)	−0.0119* (0.0069)	−0.0033 (0.0051)	−0.0118* (0.0069)	0.0014 (0.0086)	0.0004 (0.0062)	0.0206* (0.0122)
Panel B: CS DiD Short Window (2000–2009)								
High exposure estimate	—		—		—		−0.0114 (0.0109)	0.0283 (0.0201)
Panel C: Specification Details								
Estimator	Pooled OLS		One-way FE		Two-way FE		Callaway-Sant’Anna (2021)	
Treatment definition	GFC dummy × exposure		GFC dummy × exposure		GFC dummy × exposure		Formal crisis entry	
Common shock	Unabsorbed		Unabsorbed		Absorbed by year FE		Not-yet-treated controls	
Country heterogeneity	Uncontrolled		Absorbed by country FE		Absorbed by country FE		Conditional parallel trends	
Countries	131		131		130		106	

Notes: This table compares horse-race estimates across four methodologically distinct specifications. Standard errors are clustered at the country level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10 brings together the four specifications side by side, allowing the robustness checks to come to life.

The central finding is the stability of the credit channel coefficients across all specifications. Results from the pooled OLS produce a horse-race estimate of -0.0176, which is statistically significant at the 1 percent level. The Country FE horse-race estimate is -0.0119, the TWFE estimate is -0.0118, and the CS DiD high credit ATT in the short window is -0.0114. These findings are virtually identical in magnitude and direction despite each specification carrying fundamentally different identifying assumptions.

The CS DiD full window estimate of 0.0004 for the credit channel appears to break this pattern at first glance, but it does not contradict the other specifications. The full window runs through 2019, capturing a decade of post-crisis recovery during which high-credit countries recovered

faster than low-credit economies, as shown in the heterogeneity analysis. The short window estimate in Panel B tells a more comparable story - the coefficient of -0.0114 is virtually identical to the TWFE estimate of -0.0118 in both magnitude and direction. It is derived from an entirely different identification strategy using formal crisis entry rather than a GFC dummy interaction. That two approaches using fundamentally different sources of variation arrive at the same number suggests the robustness of the results.

In all four methods, the trade channel does not amplify crisis transmission. The results suggest that trade openness functions less as a shock transmission channel and more as a shock absorption mechanism, enabling economies to buffer external demand shocks rather than amplify them. Notably, the Country FE specification produces a significant trade coefficient in isolation, but as shown earlier, this disappears entirely once credit exposure is controlled for.

The specification details in Panel C showcase the underlying assumptions i.e., what each method does and does not control for. This helps understand the common shocks and the way in which country heterogeneity affects the results. Pooled OLS controls for neither. Country fixed effects absorb country heterogeneity but leave the common shock uncontrolled. TWFE absorbs both. The staggered DiD model handles the common shock through not-yet-treated comparison groups and country heterogeneity through conditional parallel trends. When using four different ways to ask the same question, the only survival remains the credit channel.

By having established the robustness of the credit channel finding across all frameworks: the country fixed effects specification, the two-way fixed effects specification, and the staggered

difference-in-differences estimator of Callaway and Sant'Anna (2021), the analysis turns to a different question. The other estimation framework, which is employed, is a probit model. This addresses a complementary yet distinct question. Instead of asking how much pre-crisis exposure led to reduced GDP growth, this model asks how pre-crisis exposures impacted the probability of a country crossing the threshold into a formal crisis. A country could experience significant reductions within its GDP growth during the GFC without meeting the criterion for a formal banking crisis - China being a prominent example. This allows the model to capture different dimensions of crisis transmission.

For this model, the sample is restricted to the GFC window of 2007 to 2009 across 119 countries. The dependent variable is a binary indicator equivalent to 1 if a country experienced a formal financial crisis in a given year, sourced from Hardy and Sever (2021) and Laeven and Valencia (2020). Standard errors are clustered at the country level. The specification takes the following form:

$$P(\text{Crisis}_{it} = 1) = \Phi (\alpha + \beta_1 \text{CreditExposure}_{i, 2006} + \beta_2 \text{TradeExposure}_{i, 2006} + X'_{it}\gamma)$$

$\text{CreditExposure}_{i, 2006}$ and $\text{TradeExposure}_{i, 2006}$ are pre-crisis exposure measures fixed at 2006 values as in the main specification, and X'_{it} is the vector of time-varying controls. Average marginal effects are reported alongside probit coefficients, as they are expressed in units of probability and are therefore more directly interpretable.

The coefficients of β_1 and β_2 are the primary variables of interest. A positive and significant β_1 and β_2 implies that countries with higher pre-crisis exposure are more likely to experience a formal financial crisis during the GFC. In case the coefficients are close to zero and insignificant,

it would imply that the channels had no systemic role in determining crisis incidence. The findings for this model are shown in table 11.

Table 11: Probit Model: Determinants of Crisis Probability (2007–2009)

	(1) Probit Coefficients	(2) Marginal Effects
<i>Panel A: Transmission Channel Variables</i>		
Pre-Crisis Credit Exposure	0.7911*** (0.2940)	0.1258*** (0.0430)
Pre-Crisis Trade Exposure	−0.2102 (0.4014)	−0.0334 (0.0642)
<i>Panel B: Control Variables</i>		
Trade Sector Share of GDP	−0.3192 (1.1371)	−0.0507 (0.1798)
Government Consumption	3.9619 (2.6330)	0.6298 (0.4095)
FDI Inflows	0.7880 (1.6096)	0.1253 (0.2588)
Mean Years of Schooling	0.1549*** (0.0525)	0.0246*** (0.0089)
Constant	−3.3949*** (0.8212)	
<i>Panel C: Model Diagnostics</i>		
Observations	355	355
Countries	119	119

Notes: Dependent variable is a binary indicator equal to one if a country experienced a financial crisis in a given year, sourced from Hardy and Sever (2021) and Laeven and Valencia (2020). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The probit results strongly reinforce and extend the main regression findings. Pre-crisis credit exposure is positive and significant at the 1 percent level in both columns: the probit coefficient is 0.791 and the average marginal effect is 0.126. The marginal effect implies that a one unit increase in the pre-crisis credit-to-GDP ratio raises the probability of experiencing a formal financial crisis during the GFC by 12.6 percentage points.

As for the trade channel, the probit results provide no evidence that it played a meaningful role in crisis incidence during the GFC. Pre-crisis trade exposure is negative and statistically insignificant in both specifications: the probit coefficient is −0.210 and the average marginal effect is −0.033. The marginal effect implies that a one unit increase in trade openness is associated with a 3.3 percentage point decrease in the probability of experiencing a formal

financial crisis; however, this estimate is highly imprecise. The 95 percent confidence interval spans values that are both negative and positive, which indicates that the data is consistent with both a modest protective effect and a modest risk-enhancing effect of trade exposure. Overall, these results reinforce the fixed-effects findings and provide additional evidence that trade openness did not systematically influence crisis transmission during the GFC.

As for the control variables, mean years of schooling is positive and significant. This indicates that countries with higher levels of education were more likely to experience formal financial crises during the GFC. This could be related to these countries having advanced economies as well as deep financial systems and extensive cross-border financial systems that link each other.

The other control variables such as government consumption, FDI inflows, and the tradeable sector, however, were all statistically insignificant. The government consumption is interesting to note - since within the fixed effects model, the variable was significant. Moreover, independent of the credit and trade exposure channels, the tradeable sector was also insignificant in the probit specification.

VI. Policy Implications

The findings discussed above extend beyond their empirical nature; they carry insights for policymakers on ways to create safeguards for future crisis prevention. Given its dominance in the horse-race specification, pre-crisis credit depth is the dominant link for transmission crises. This means that resilience frameworks built around trade exposure are deemed to provide less buffer.

In terms of macroprudential regulation, the credit channel reflects a concrete micro-level dynamic. The failure of financial institutions, where initial losses lead to falling risk-bearing capacity and forced liquidation, spreads distress across systems independent of the trade relationship (Hasman & Samartín, 2025; Boeck & Mori, 2025). Thus, behind the wheel of the financial accelerator lies the wealth effect. As credit expands, the wealth effect operates very precisely. To act as a counterbalance, capital buffer requirements need to be redefined to target this vulnerability by obliging banks to accumulate capital during expansions, which limits the fuel available to the financial accelerator. This, thereby, limits the amplification of shocks when conditions deteriorate. As seen before, according to the results from the full sample, a one standard deviation increase in pre-crisis credit exposure implies a drop in total GDP output of 0.48 percentage points.

For emerging market economies, the stakes seem to be higher. Their growth from 20 to 40 percent of world economic activity in the recent decade and the findings of this thesis suggest heightened sensitivity to international credit shocks. The Dabla-Norris and Bal Gündüz (2012)

finding that institutional quality and policy buffers shape vulnerability in low-income countries complements this paper's credit channel evidence rather than contradicting it (Yuan et al., 2023). Taken together, they suggest that a complete vulnerability assessment needs to weigh both financial depth and structural buffers depending on where a country sits in the development spectrum.

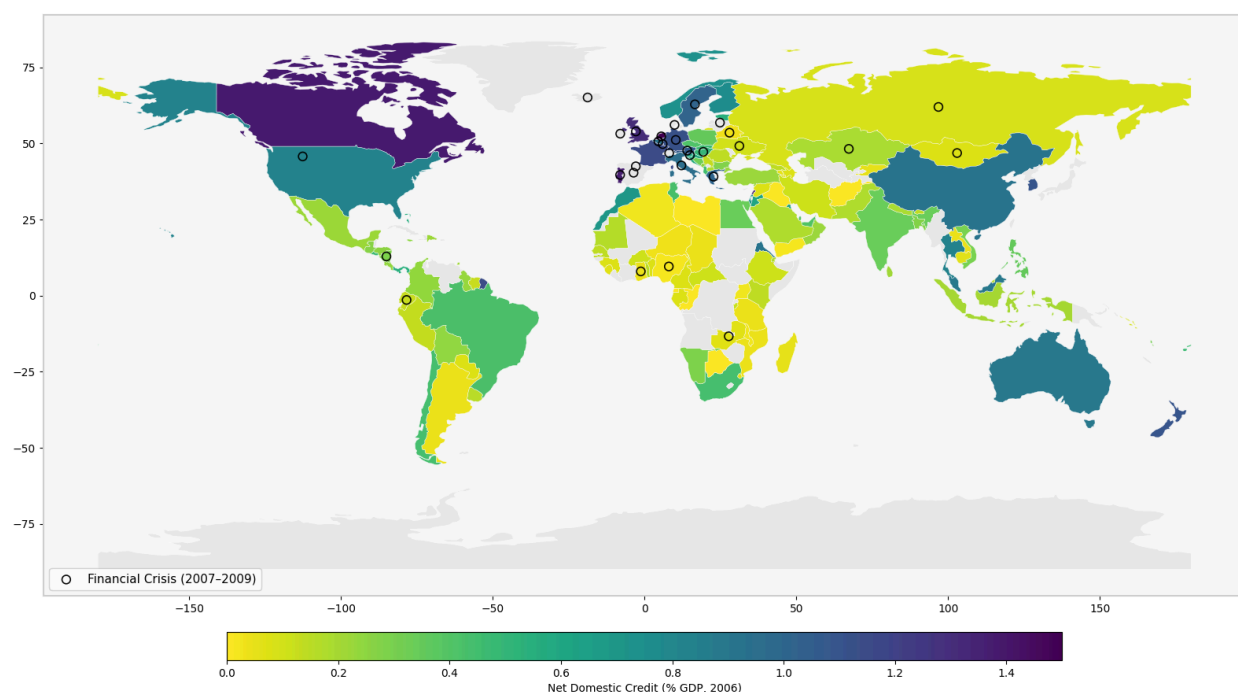
The staggered DiD framework suggests that trade openness can play a protective role, which merits its own policy implications. High-trade countries experienced relatively smaller contractions during the GFC, which may be explained by the fact that it is not trade itself but what tends to accompany it. With trade integration comes the ability to accumulate stronger buffers over time, which allows domestic institutions to act as lenders of last resort and thereby protect the country from foreign reserve crises (Hasman & Samartín, 2025). As a policy lesson, it becomes imperative to focus on systems that create effective management of reserves rather than just trade itself.

A note of caution remains, however: while this thesis emphasises the importance of the credit channel during the GFC, it should be taken with a grain of salt. The credit channel dominated in 2008. It does not mean that it will always dominate. As the global economy continues to expand and integrate, the channel hierarchy identified may shift over time. Thus, policymakers designing resilience frameworks for the next crisis should monitor both the accumulation of domestic credit and the deepening of trade linkages.

Appendix 1. Credit Exposure and Financial Crises Incidence

To further examine the role of financial conditions in crisis vulnerability, this section isolates the credit channel. This helps to examine the relationship between pre-crisis domestic credit exposure and the incidence of financial crises during the 2007–2009 period.

Figure 3. Pre-Crisis Credit Exposure and Incidence of Financial Crises



Notes: Countries where the onset of the crisis took place are marked with a black circle

Figure 3 plots pre-crisis domestic credit as a proportion of the GDP against crisis incidence. The underlying colors reflect the level of pre-crisis credit exposure.

With this visual, it becomes clear that countries that experienced a financial crisis tend to exhibit higher pre-crisis credit exposure relative to non-crisis economies. This evidence is consistent with the regression results presented in the main text, reinforcing the interpretation that pre-crisis financial deepening amplified vulnerability during the global financial crisis.

Appendix 2. Heterogeneity in Transmission Channels Across Development Levels

This appendix explores whether the relative importance of credit and trade transmission channels varies systematically with countries' level of development. While the main analysis establishes that the credit channel dominates on average, across the full sample, differences in economic structure may imply heterogeneous exposure across country groups that averages can obscure.

The main analysis established that advanced economies entered the GFC with much higher pre-crisis credit exposure as opposed to developing countries. For the financial accelerator to operate, it is necessary to have deep credit markets. Therefore, its grip may be weaker in the developing world where credit intermediation is shallow and financial systems are less integrated.

Table 12 tests this by estimating the baseline horse-race specification using TWFE for both developed and developing economies. Within the table, the first column reproduces the results from table 6 for the horse-race specification; whereas, columns 2 and 3 further divide it based on the category of the economy.

Table 12: TWFE - Developed vs. Developing Countries

	(1) Full Sample	(2) Developed	(3) Developing
<i>Panel A: Key Variables of Interest</i>			
GFC $\times$ Credit Exposure	-0.0118* (0.0069)	0.0043 (0.0076)	-0.0059 (0.0188)
GFC $\times$ Trade Exposure	0.0014 (0.0086)	-0.0107 (0.0088)	0.0029 (0.0110)
<i>Panel B: Control Variables</i>			
Government Consumption	-0.3448*** (0.1148)	-0.8260*** (0.1775)	-0.3129** (0.1253)
<i>Panel C: Model Diagnostics</i>			
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	2,435	418	2,017
Countries	131	22	109
Within R^2	0.109	0.558	0.096

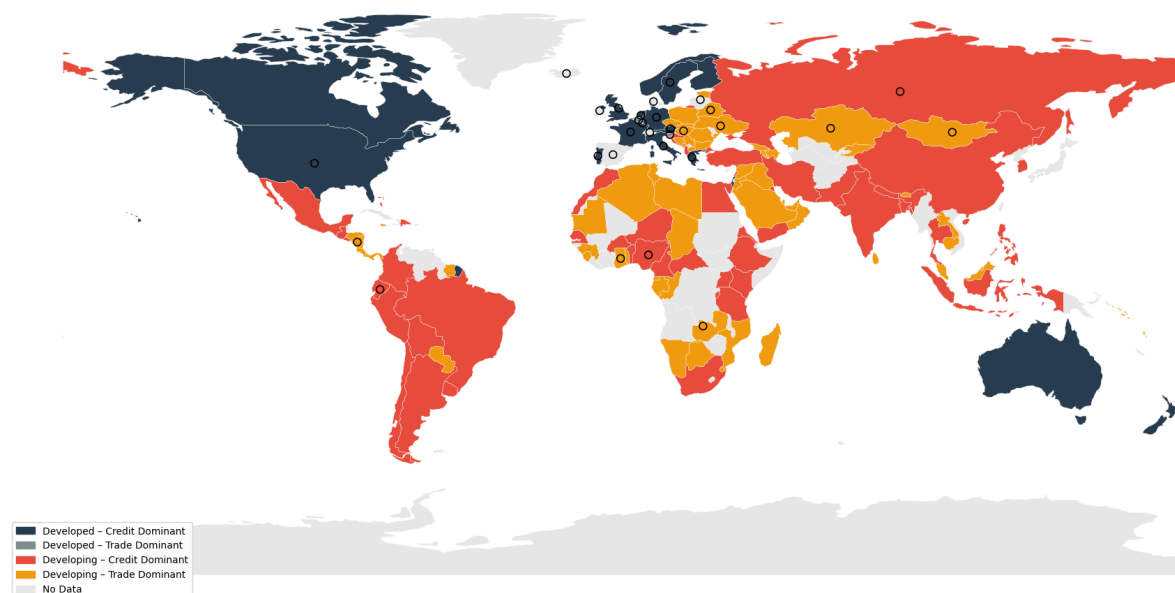
Notes: Dependent variable is GDP growth, measured as the log difference of real GDP at constant 2015 prices. All models estimated using two-way fixed effects (country and year) with standard errors clustered at the country level, reported in parentheses. Country classification follows the IMF Advanced Economies list.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

As seen within the subsamples, the coefficients lose statistical significance. This reflects the substantially reduced sample sizes particularly for developing countries, which have a subsample of only 22 countries, rather than the absence of an effect. The R squared for developed economies further suggests that year fixed effects explain a greater share of GDP growth variation in advanced economies.

To complement the regression evidence, Figure 4 provides a geographic visualization of the dominant pre-crisis transmission channel for each country, using the pre-2007 structure.

Figure 4. Development Status and Dominant Transmission Channel



Notes: Countries where the onset of the crisis took place are marked with a black circle; whereas different colours reflect their development status and the dominant channel of transmission.

Figure 4 brings this heterogeneity to life. The map classifies each country by development status and dominant transmission channel, which is defined by whether standardised pre-crisis credit exposure exceeds standardised trade exposure. These measures are transformed into z-scores to ensure that differences in scale between the two variables do not drive the classification. The two measures are then compared to each other to find out which channel dominates:

1. $z_{\text{credit}} > z_{\text{trade}}$ for the dominance of the credit channel;
2. $z_{\text{trade}} > z_{\text{credit}}$ for the dominance of the trade channel

A clear pattern emerges: advanced economies are predominantly credit-driven, reflecting deeper financial integration and stronger banking-sector linkages. On the other hand, the developing

world tells a nuanced story, exhibiting greater heterogeneity. A substantial share of these countries across Sub-Saharan Africa, South and Southeast Asia, and parts of Eastern Europe appear trade-dominant, implying that trade openness is a more prominent feature of their integration into the global economy than financial depth.

Taken together, these results suggest that while credit conditions remain the primary mechanism of crisis transmission globally, trade linkages may play a more prominent role in shaping vulnerability among developing economies.

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